

Assignment

1. Explain the components of embedded hardware? Explain how washing machine is an embedded system
2. What is single purpose processor? What are the benefits of choosing a single purpose processor over a general purpose processor?
3. Distinguish between a combinational and sequential logic?
4. Design a 3-bit counter that counts the following sequence 1, 2, 4, 5, 7, 1, 2,... etc. This counter has an output “odd” whose value is 1 when the current count value is odd. Use the sequential design technique. Start from a state diagram, draw the truth table, minimize the design logic and draw the final circuit.
5. Design a soda machine controller, given that a soda cost 75 cent and your machine accepts quarters only. Draw a black-box view, come up with a state diagram and state table, minimize the logic, and then draw the final circuit
6. How does the programmer view a microcontroller/ what is necessary knowledge one should have before programming the microcontroller?
7. What is pipelining? What is pipelining hazards?
8. Explain Superscalar and VLIW architecture?
9. Acquire the data book of a 8051 microcontroller and list the features of 8051? Explain its different types of instruction set special function register, I/O facilities interrupt and available program and data memory.
10. Define RTOS? Explain the features of RTOS.
11. Define task? Explain task states with the help of FSM
12. What is semaphore? List the types of semaphores
13. What is task synchronization? How a common resource is shared among different application with the help of semaphore?
14. What is message queue and pipes. What are their uses?
15. Explain pre-emptive and round robin scheduling with the help of diagram.
16. Explain event register.
17. What is memory allocation? How does malloc and free function works?
18. What is memory fragmentation and memory leakage?
19. Why memory management is important in embedded system. How do you plan to manage memory in embedded system products which have so many constraints.

20. What is process? Differentiate between thread and process.
21. What is cross compilation and native compilation?
22. Illustrate how a big application is developed in embedded system?
23. List various methods being used in embedded system for debugging and testing of application as well as hardware.